

Fabio Cavagna

PhD Candidate in Aerospace Engineering

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EXPERIENCE

Siemens Digital Industries Software

Research Test and Simulation Engineer

Leuven, Belgium

Dec. 2024 – Present

- Conducted experimental tests of structures mounted on shaker tables, post-processing the acquired data to extract modal parameters with Simcenter Testlab.
- Performed numerical simulations of structures using Simcenter Nastran, validating the results against experimental data.

CNES

Structural Dynamics Intern

Paris, France

Apr. 2024 – Oct. 2024

- Defined testing procedures for the experimental campaign, including the finalization of the installation of a small-scale tank and the definition of safety measures.
- Performed numerical simulations to compute the sloshing and hydro-elastic modes of liquid tanks.

Thales Alenia Space

Master Thesis Intern

Turin, Italy

Mar. 2023 – Sept. 2023

- Conducted a comprehensive assessment of methodologies used to evaluate and compare the severity of mechanical loads, including Shock Response Spectrum, Extreme Response Spectrum, and Fatigue Damage Spectrum.
- Developed a novel approach for comparing the severity of mechanical Multi Degree of Freedom loads using MATLAB, emphasizing the analysis of interface forces in the presence of 6 DoF loads.

Polispace

Structural Engineer

Milan, Italy

Mar. 2023 – Sept. 2023

- Student association developing a CubeSat for ESA's "Fly Your Satellite!" program. Simplified CAD geometries using SolidWorks and Abaqus, integrating them into the FEM model.
- Contributed to defining structural requirements for the Thermal Vacuum Chamber Test to ensure CubeSat resilience in extreme conditions.

EDUCATION

Politecnico di Milano

PhD Candidate in Aerospace Engineering

Milan, Italy

Dec. 2024 – Present

Research topic: "Decoupling techniques based on substructuring for fixed-base analysis."

ISAE-SUPAERO

Advanced Master in Aeronautical and Space Structures

Toulouse, France

Oct. 2023 – Oct. 2024

- Team Innovative Management for Evolved Strategies – TRL analysis for LH2 propellant transitioning from space to aeronautics. Defined WBS, Gantt chart, and risk matrix.
- Space Structures – Sizing of the TMI Wall of SPOT 4 for quasi-static loads, thermal loads, and buckling through Patran/MSC Nastran.

Politecnico di Milano

M.S. in Space Engineering

Milan, Italy

Sept. 2021 – Oct. 2023

- Analysis and Testing of Space Structures – Structural compliance study of an axial-symmetric spacecraft with Vega-C. Static, modal, thermoelastic, buckling, sine test, coupled load, and acoustic analyses in MSC Nastran (ECSS qualification levels).
- Aerospace Technologies and Materials – a) Modeled a composite panel with embedded piezoelectric plate; validated natural frequencies via NX Nastran and testing. b) Optimized an additive manufacturing component with Fusion 360 and Abaqus, achieving 61.63% volume reduction.

Politecnico di Milano

B.S. in Aerospace Engineering

Milan, Italy

Sept. 2018 – Sept. 2021

SOFTWARE SKILLS

CAE: Simcenter 3D, Abaqus, Femap, MSC Nastran, Patran, Ansys Mechanical, HyperMesh

CAD: CATIA, SolidWorks, Solid Edge, Autodesk Inventor

Programming: MATLAB, Python, LaTeX

Office: Microsoft Office

LANGUAGES

Italian: Mother tongue

English: Proficient user

French: Independent user